Computer Programming (CSN11101)
ASSIGNMENT-1

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Q1. Write a program that swaps the values of two variables:

a) Using a third variable for temporary storage.
b) Without using a third variable.

A1. (a)
CODE:

```
#include<stdio.h>

int main(){
  int a,b;
  printf("Enter the first value");
  scanf("%d",&a);
  printf("Enter the second value");
  scanf("%d",&b);
  int c=b;
  b=a;
  a=c;
  printf("a : %d \n",a);
  printf("b : %d \n",b);
  return 0;
  }
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p2
Enter the first value12
Enter the second value13
a : 13
b : 12
```

(b) CODE:

```
#include<stdio.h>

int main(){
  int a,b;
  printf("Enter the first value");
  scanf("%d",&a);
  printf("Enter the second value");
  scanf("%d",&b);
  a=a+b;
  b=a-b;
  a=a-b;
  printf("a : %d \n",a);
  printf("b : %d \n",b);
  return 0;
  }
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p3
Enter the first value12
Enter the second value13
a : 13
b : 12
```

Q2. Write a program that checks if a given integer is odd or even.
A2. CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("Enter the number : ");
  scanf("%d",&n);
  if(n%2==0){printf("The number is even \n");}
  else{printf("The number is odd \n");}
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p4
Enter the number : 12
The number is even
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p4
Enter the number : 13
The number is odd
```

Q3. Write a program to find if a given number is zero, positive or negative.
A3. CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("Enter the number : ");
  scanf("%d",&n);
  if(n==0){printf("The number is zero \n");}
  else if(n>0){printf("The number is positive\n");}
  else{printf("The number is negative \n");}
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p5
Enter the number : 13
The number is positive
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p5
Enter the number : -12
The number is negative
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p5
Enter the number : 0
The number is zero
```

Q4.Write a C program to accept two integers and check whether they are equal or not
A4. CODE :

```
#include<stdio.h>

int main(){
  int a;int b;
  printf("Enter the first number : ");
  scanf("%d",&a);
  printf("Enter the second number : ");
  scanf("%d",&b);
  if(a==b){printf("Both the numbers are equal \n");}
  else{printf("Both the numbers are not equal \n");}
}
```

OUTPUT :

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p6
Enter the first number : 13
Enter the second number : 12
Both the numbers are not equal
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p6
Enter the first number : 12
Enter the second number : 12
Both the numbers are equal
```

Q5. Write a C program to check whether an alphabet is a vowel or a consonant.
A5. CODE :

```
#include<stdio.h>

int main(){
  char a;
  printf("Enter the alphabet : ");
  scanf("%c",&a);
  if(a=='a' || a=='e' || a=='i' || a=='o' || a=='u'){printf("It is a vowel \n");}
  else{printf("It is a consonant \n");}
}
```

OUTPUT :

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p7
Enter the alphabet : a
It is a vowel
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p7
Enter the alphabet : b
It is a consonant
```

Q6.Write a program that takes three integers and determines the largest among them.
A6. CODE:

```
#include <stdio.h>

int main(){
  int a;int b;int c;
  printf("Enter the first number : ");
  scanf("%d",&a);
  printf("Enter the second number : ");
  scanf("%d",&b);
  printf("Enter the third number : ");
  scanf("%d",&c);
  if(a>=b && a>=c){printf("first number is the largest \n");}
  else if(b>=c && b>=a){printf("Second number is the largest \n");}
  else{printf("Third number is the largest \n");}
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p8
Enter the first number : 12
Enter the second number : 12
Enter the third number : 11
first number is the largest
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p8
Enter the first number : 10
Enter the second number : 11
Enter the third number : 12
Third number is the largest
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004$ ./p8
Enter the first number : 10
Enter the second number : 15
Enter the third number : 12
Second number is the largest
```